Shenyu Qin

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Education

Shanghai Jiao Tong University *Sept. 2022 – Present*
Undergraduate in Computer Science

- GPA: 3.75/4.3 (88.36/100)
- Member of John Hopcroft Honors Class, a CS program at Zhiyuan College for the top 10% of students, with a focus on Theoretical Computer Science
- Relevant Coursework: Mathematical Analysis II (93), Combinatorial Mathematics (91), Probability Theory (91), Computational Complexity (98), Online Algorithms (95), Statistical Learning Theory (97)

Experience

Existence of α-strong and α-dominating committees in Impartial Culture *DIMACS, Rutgers University*
Aug. 2025 – Present
Advised by Prof. Kangning Wang and Prof. Lirong Xia

- Studied the existence of α-strong (equivalently, $(1-\alpha)$-undominated) and α-dominating committees of size k in an impartial culture when the number of candidates m is much larger than the number of voters n
- Established two sharp thresholds for α: an α-strong committee exists w.h.p. iff $\alpha < \frac{1-k}{k}$, and an α-dominating committee exists w.h.p. iff $\alpha < \frac{1-k}{2k}$, when $m \gg n \gg 0$
- Used probabilistic methods including Chernoff bounds and union bounds to analyze the probability of such committees' existence

Online Load and Graph Balancing in Random Order Model *John Hopcroft Center, SJTU*
Advised by Prof. Yuhao Zhang *Jul. 2024 – Mar. 2025*

- Studied the online load balancing problem in random arrival order model, aiming to close the gap between the lower bound of $\Omega(\sqrt{\log m})$ and the upper bound of $O(\log m/\log\log m)$ by designing an $O(\sqrt{\log m})$-competitive algorithm, where m is the number of machines
- Investigated the online graph balancing problem, which is a special case of the original problem, trying to analyze the competitive ratio of a newly-designed algorithm in this case (stuck on this part)
- Participated in TCS Reading Group every week

Projects

Housing Price Prediction *Jul. 2024*
Course Project in Data Mining

- Developed a data mining model to analyze the relationship between housing prices and other variables using a partial dataset of Airbnb listings in New York City
- Predicted the price of a given listing and participated in a Kaggle competition

Towards the chat among distinct LLM agents *May. 2024*
Course Project in Natural Language Processing and Large Language Model

- Investigated the collaborative potential of completely distinct LLM agents (GPT, Llama, and Wenxin Yiyan) through structured dialogues and wrote a paper about the results
- Discovered a new technique termed "Virtual Conversation", where an LLM internally simulates a multi-agent discussion

Lexical and Syntax Analysis of a Programming Language with Array and String Types *Nov. 2023*
Course Project in Programming Languages Design and Implementation

- Performed lexical and syntax analysis on the WHILE language with array and string types, and implemented syntax tree output
- Tools Used: Bison, Flex

Deque *Apr. 2023*

Course Project in Program Design and Data Structure

- Implemented deque in my own way with arrays and a doubly linked list, meeting the complexity requirement

Honors and Awards

Zhiyuan Honors Scholarship (Top 10%), SJTU *2022, 2023, 2024*

Skills

Programming Languages: C++, C, Python, Rust

Software: LaTeX, MATLAB, Coq

Languages: Mandarin(Native), English(TOEFL: 104)